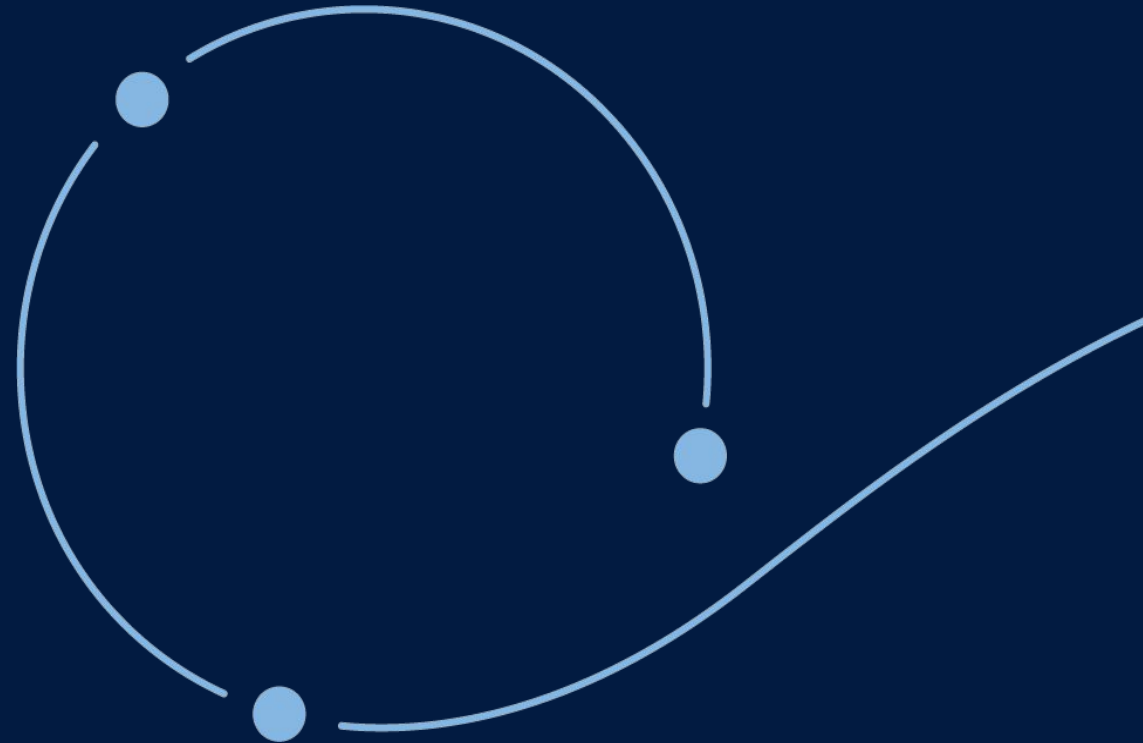


Pushing the limits of latent fingerprint identification with synthetic data

IFPC 2025 Conference

Igor Jánoš
Head of R&D

 INNOVATRICS

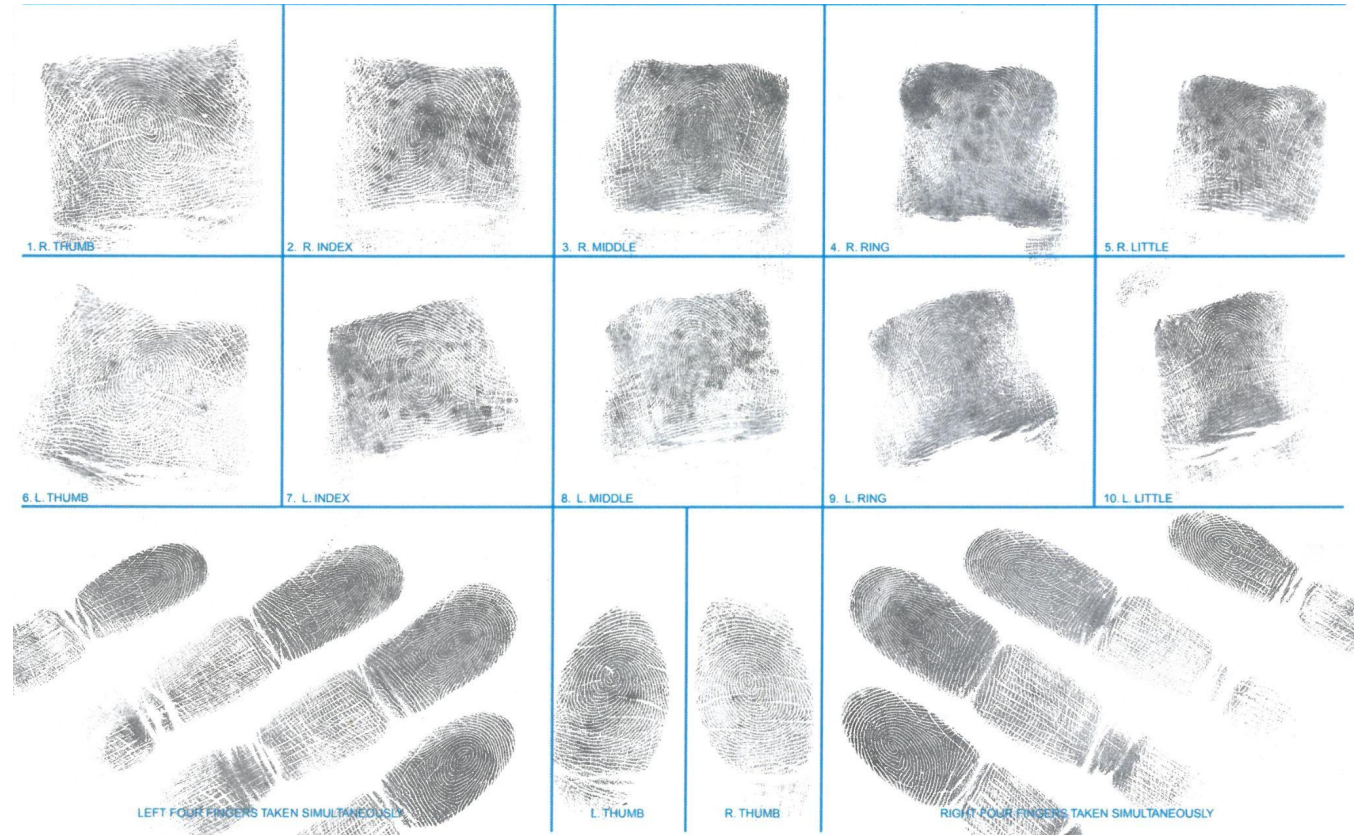


Motivation

- Images of latent fingerprints are scarce
- Often they are confidential
 - part of an investigation (ongoing or closed)
- Matching pairs (latent - plain) are difficult to get

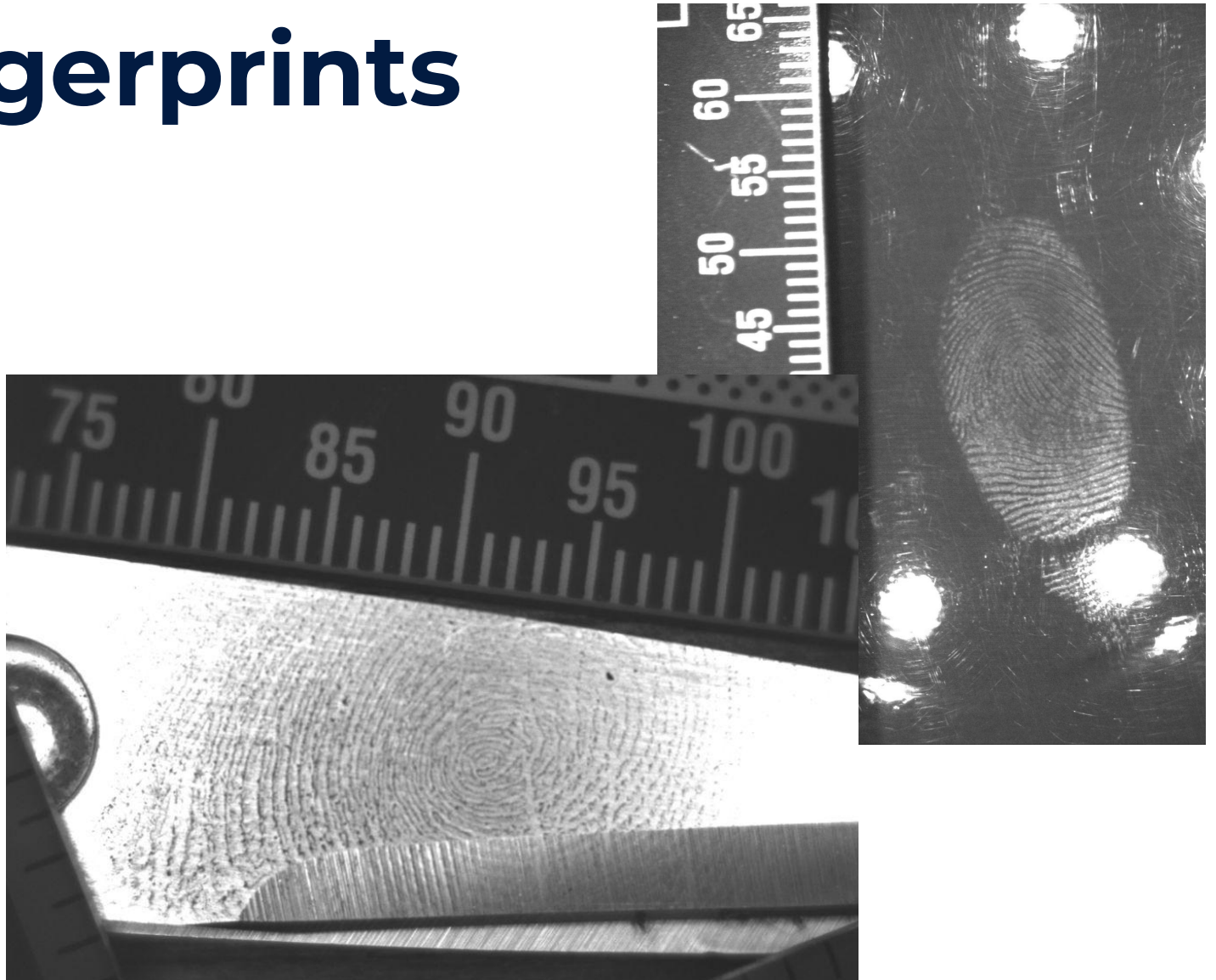
Known Fingerprints

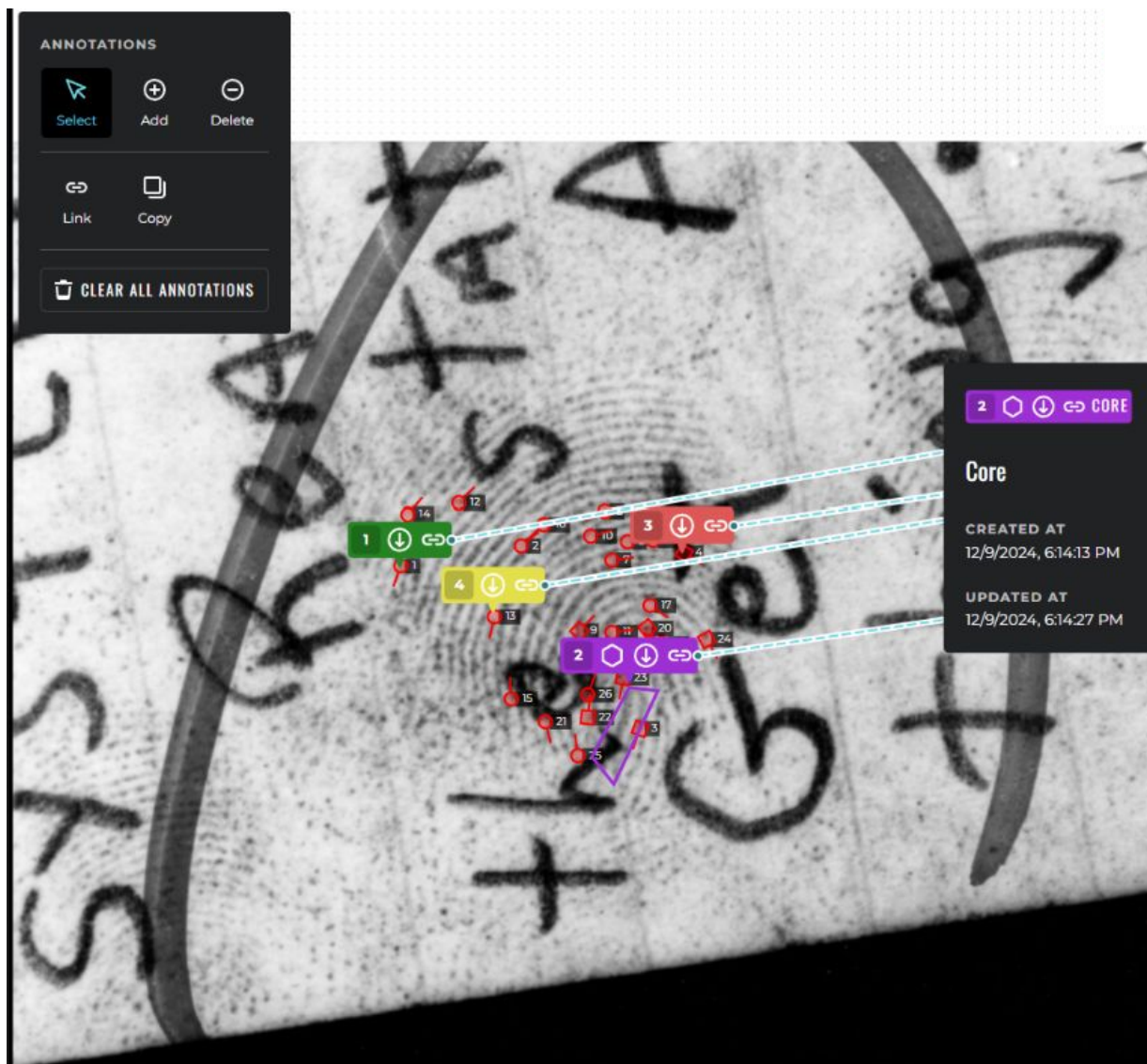
- Acquired using ink or digital scanners
- Controlled environment
- Constant DPI
- Less diverse



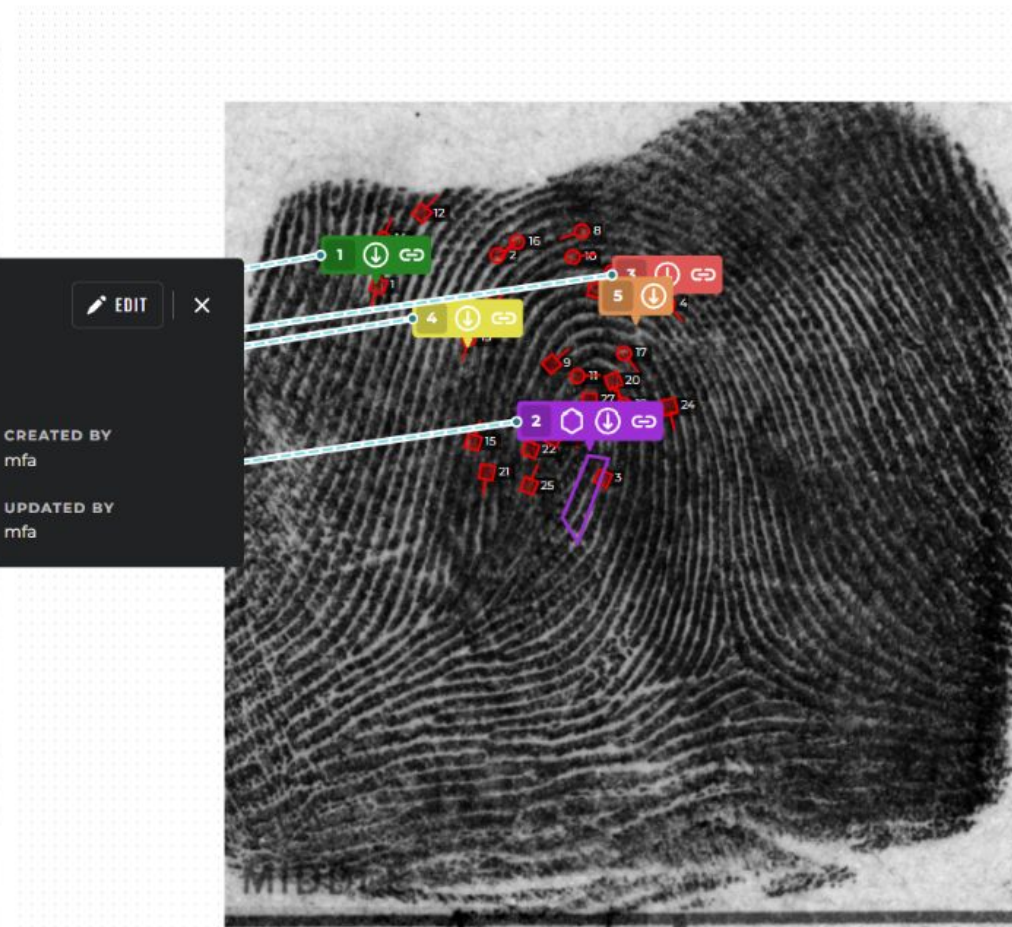
Unknown Fingerprints

- Diverse backgrounds
- Partial information
- Acquired using different techniques

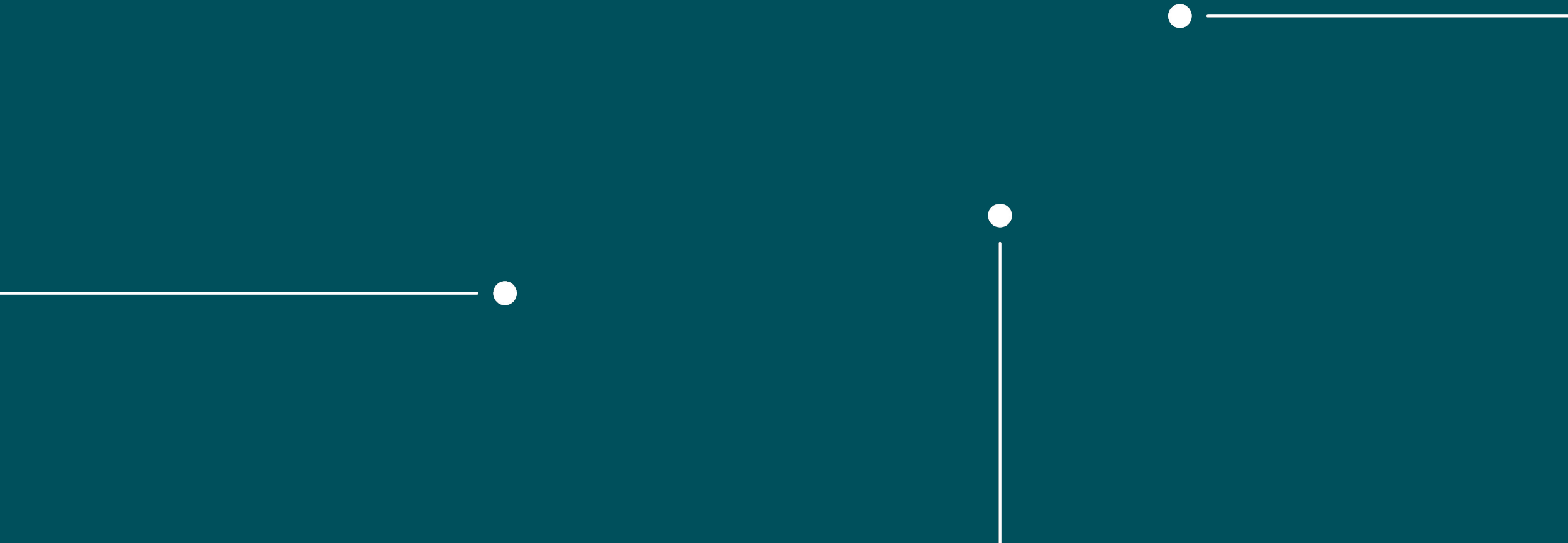




- Annotating latent fingerprints is difficult
- Having forensic experts do the annotation is expensive
- Laymen will not always be consistent



Our Approach

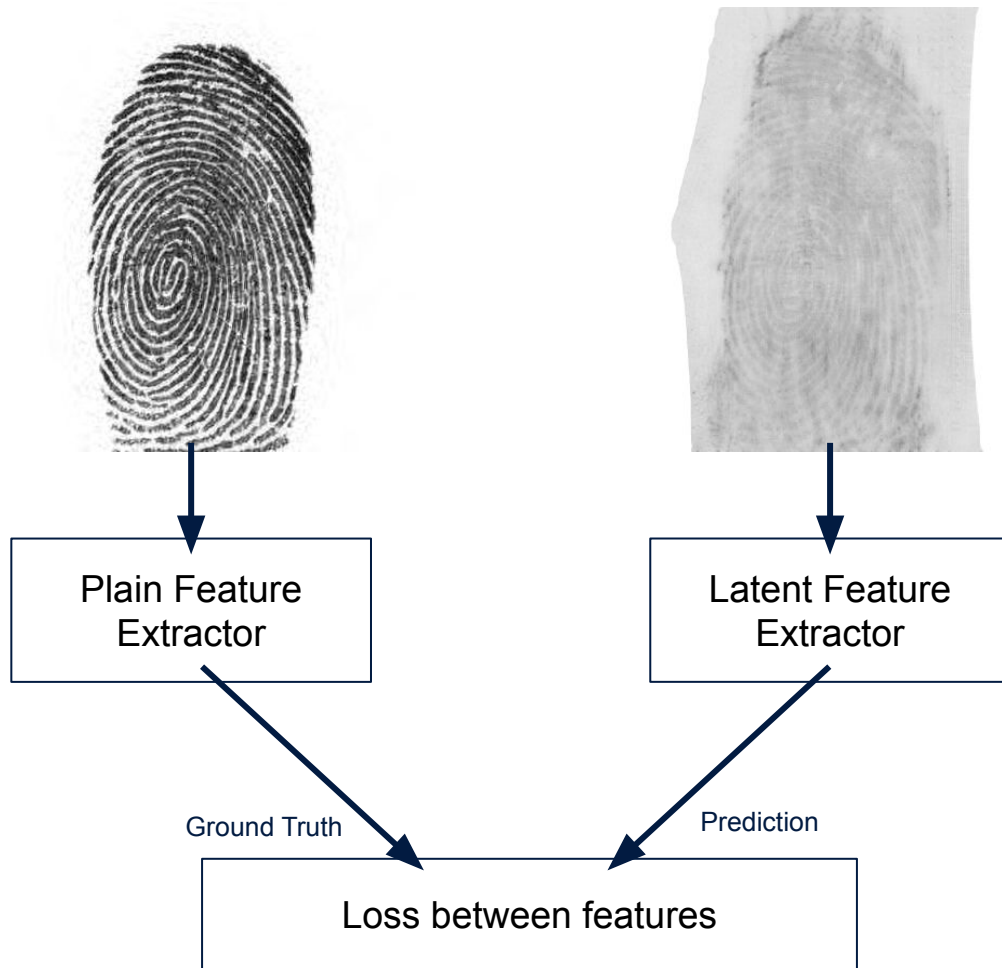


Our Approach

- **Train on purely synthetic latent images**
- We acknowledge lot of research on synthetic fingerprint generation
 - Cycle-GAN-based approaches
 - Diffusion-based approaches
 - Procedural approaches

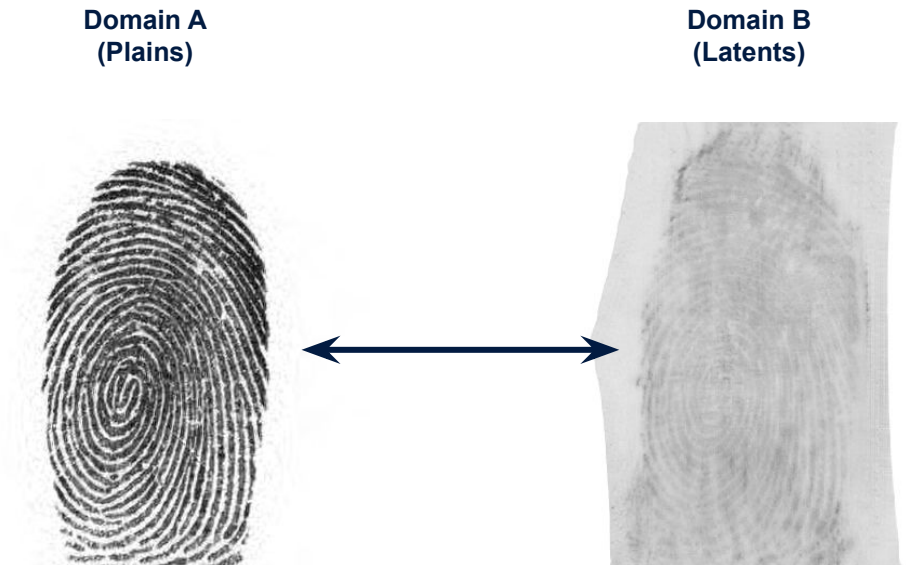
Training

- Predict the same features from plain and latent images



Training Data

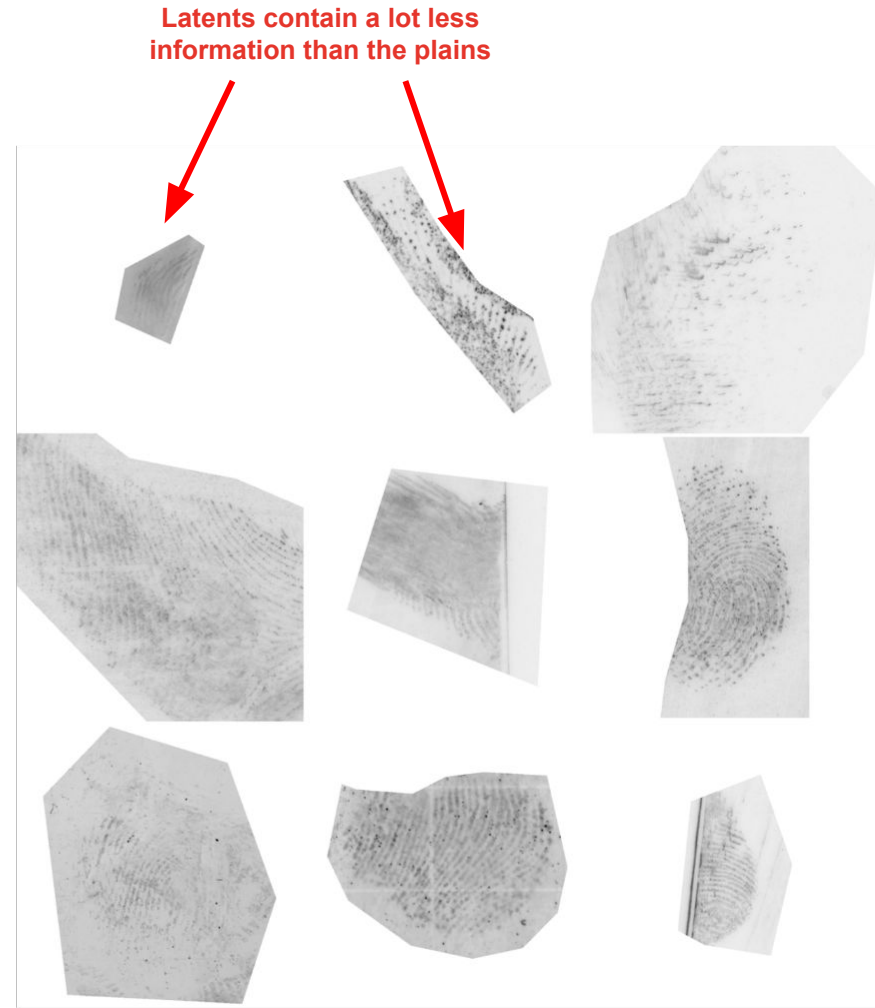
- Produce big bag of image pairs
 - Augmentation
 - GANs
 - Diffusion
- We have no real matching pairs
 - Unpaired image translation



Training Data



Real Plains

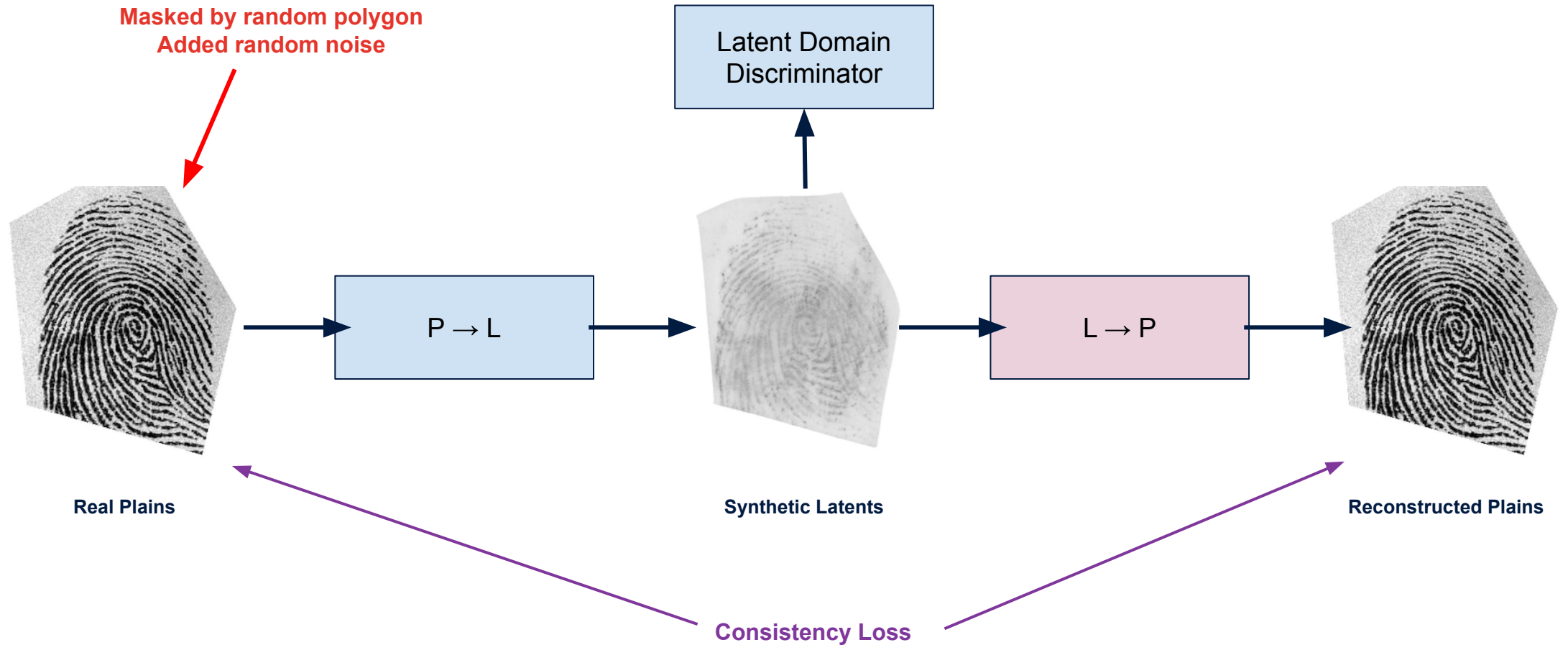


Real Latents (SD302)

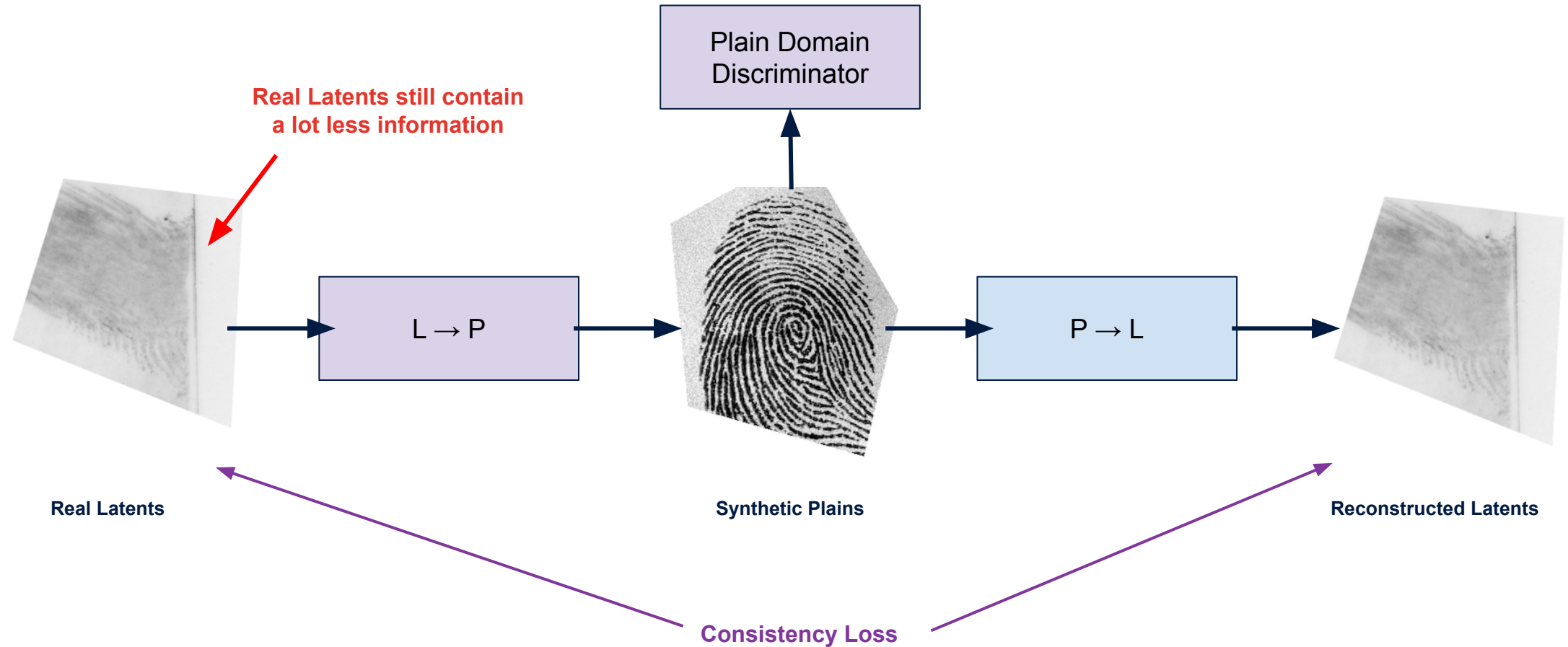
Submission 1



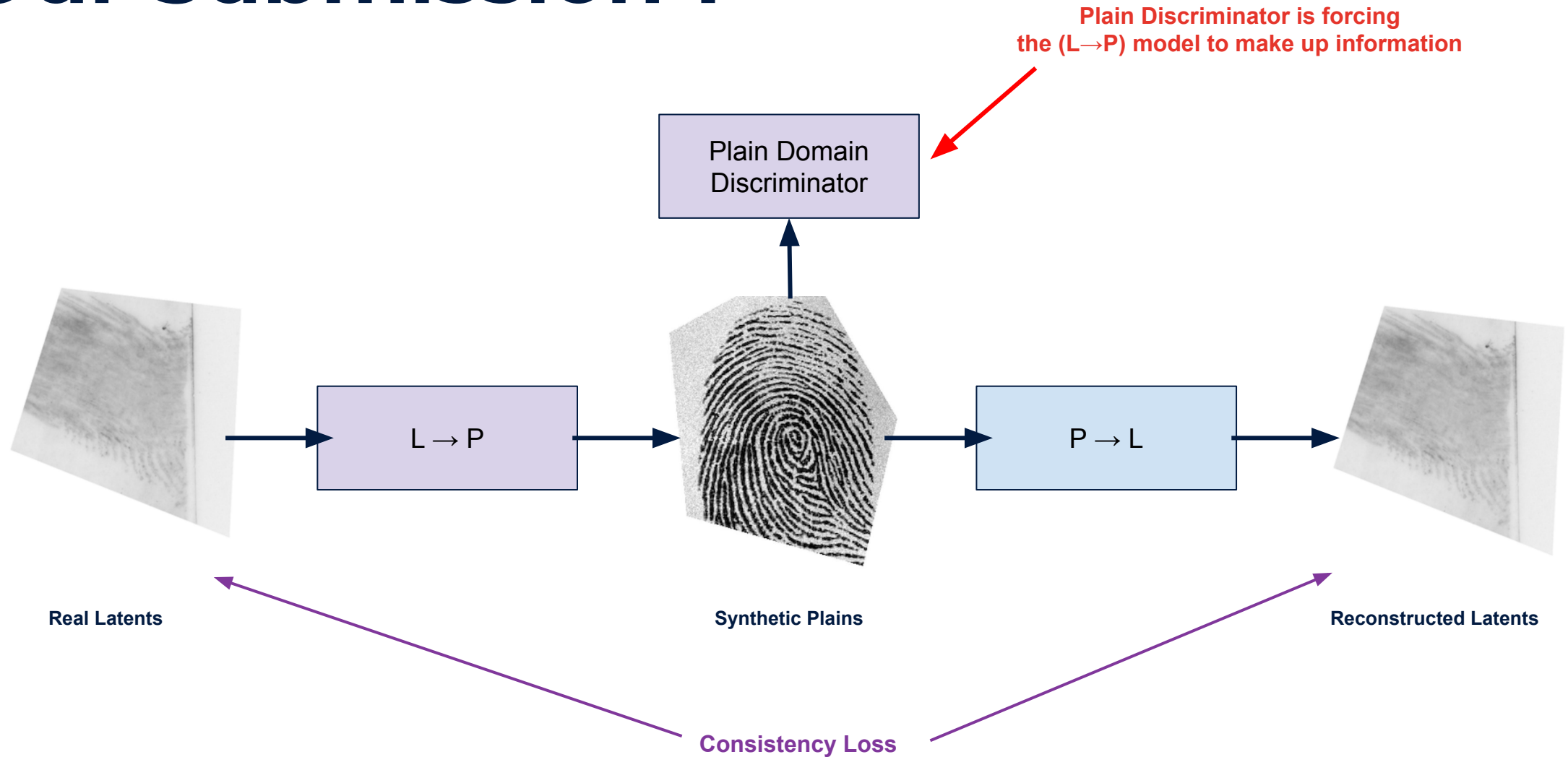
Our Submission 1



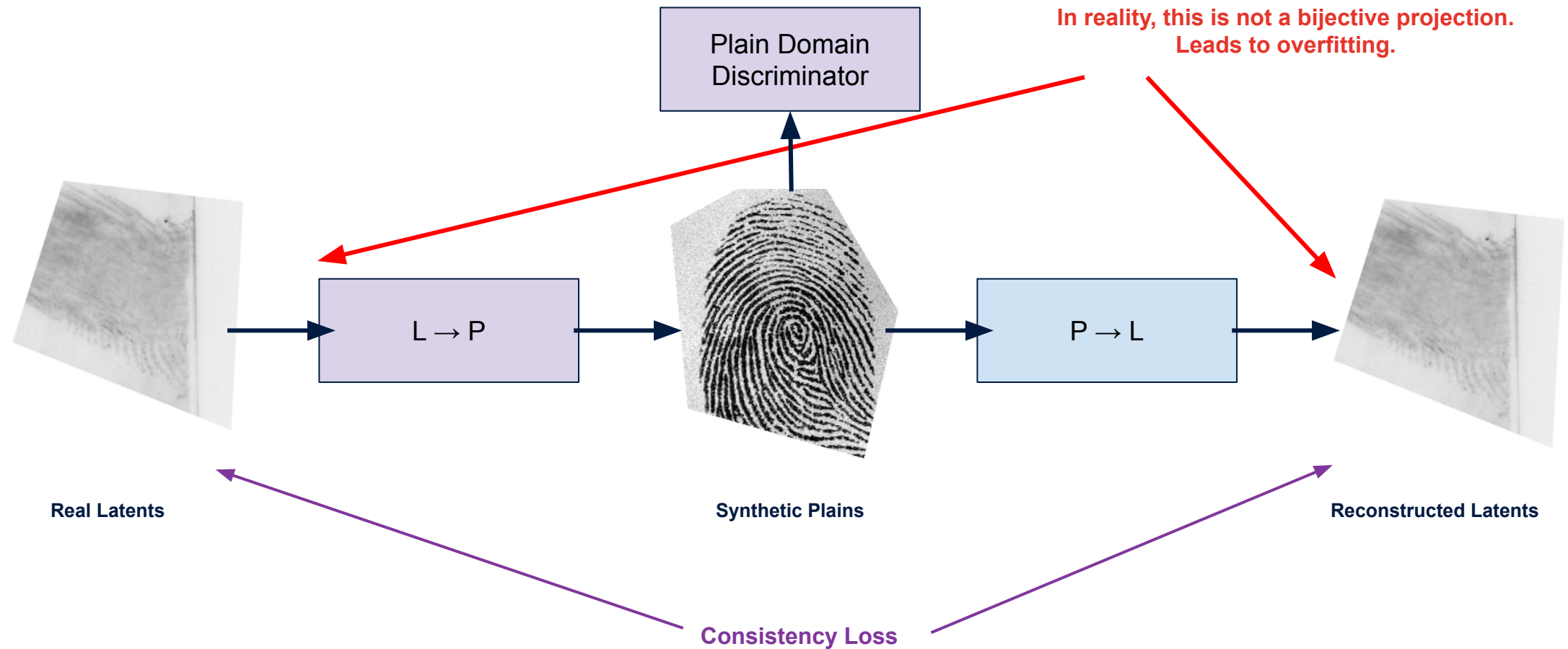
Our Submission 1

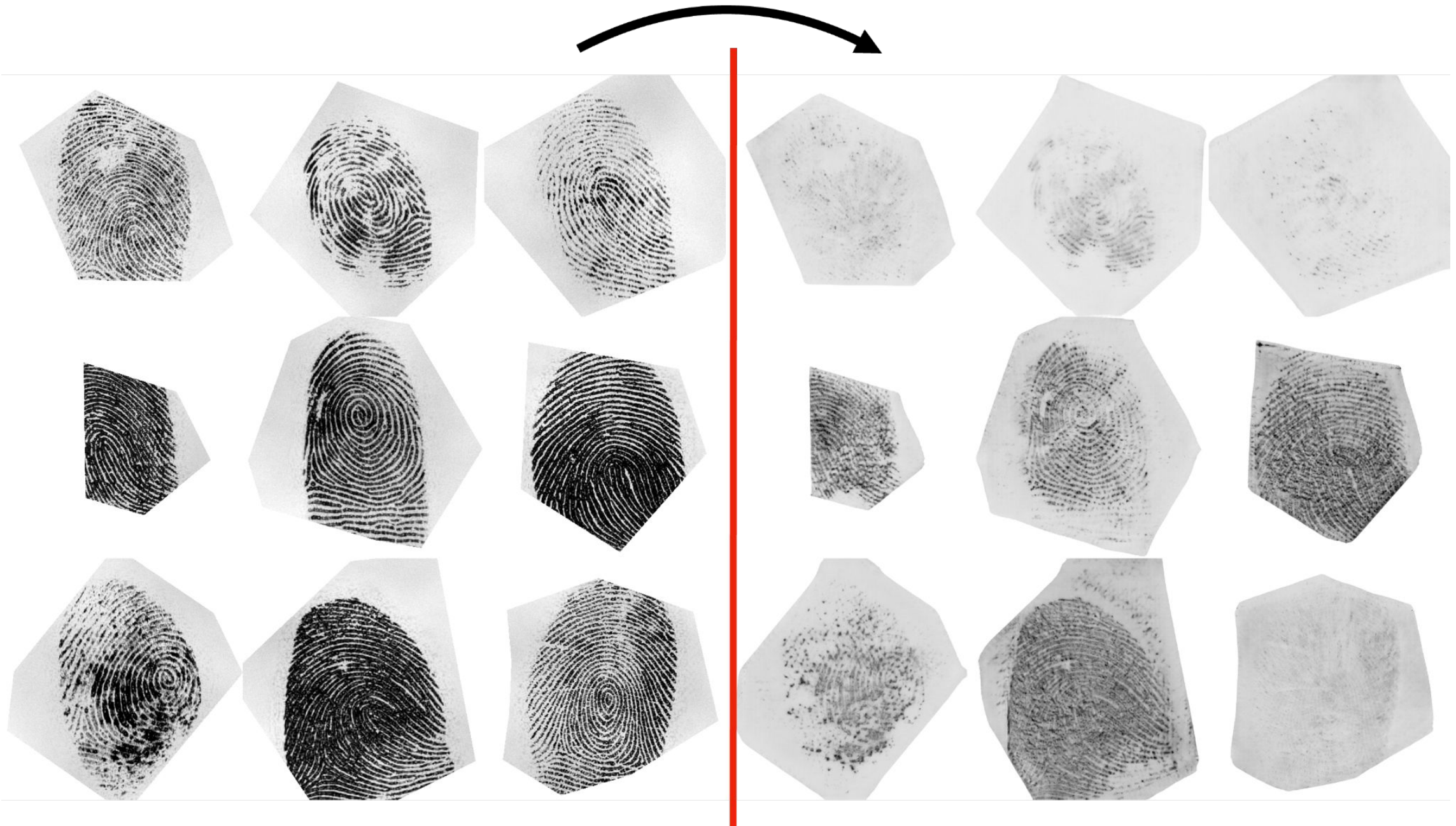


Our Submission 1



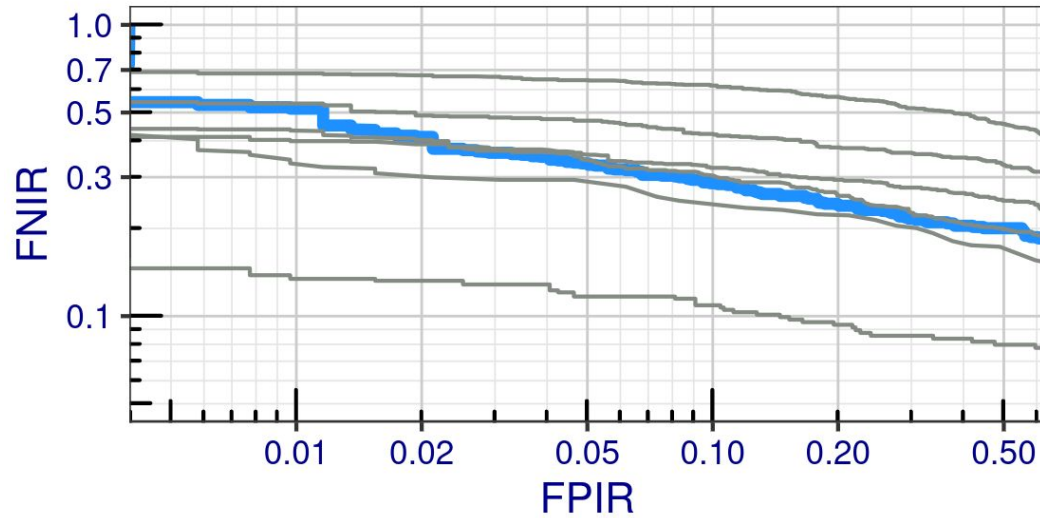
Our Submission 1



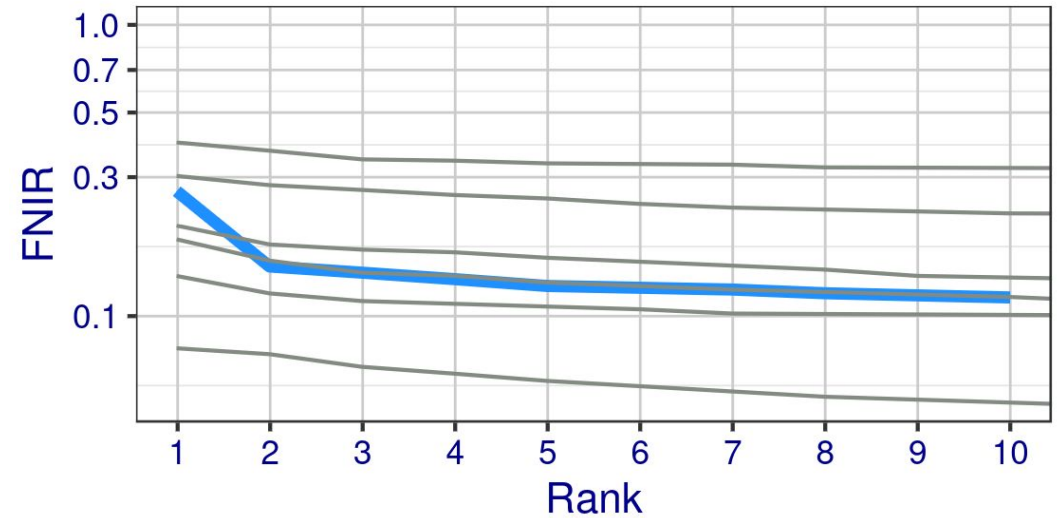


Our Submission 1

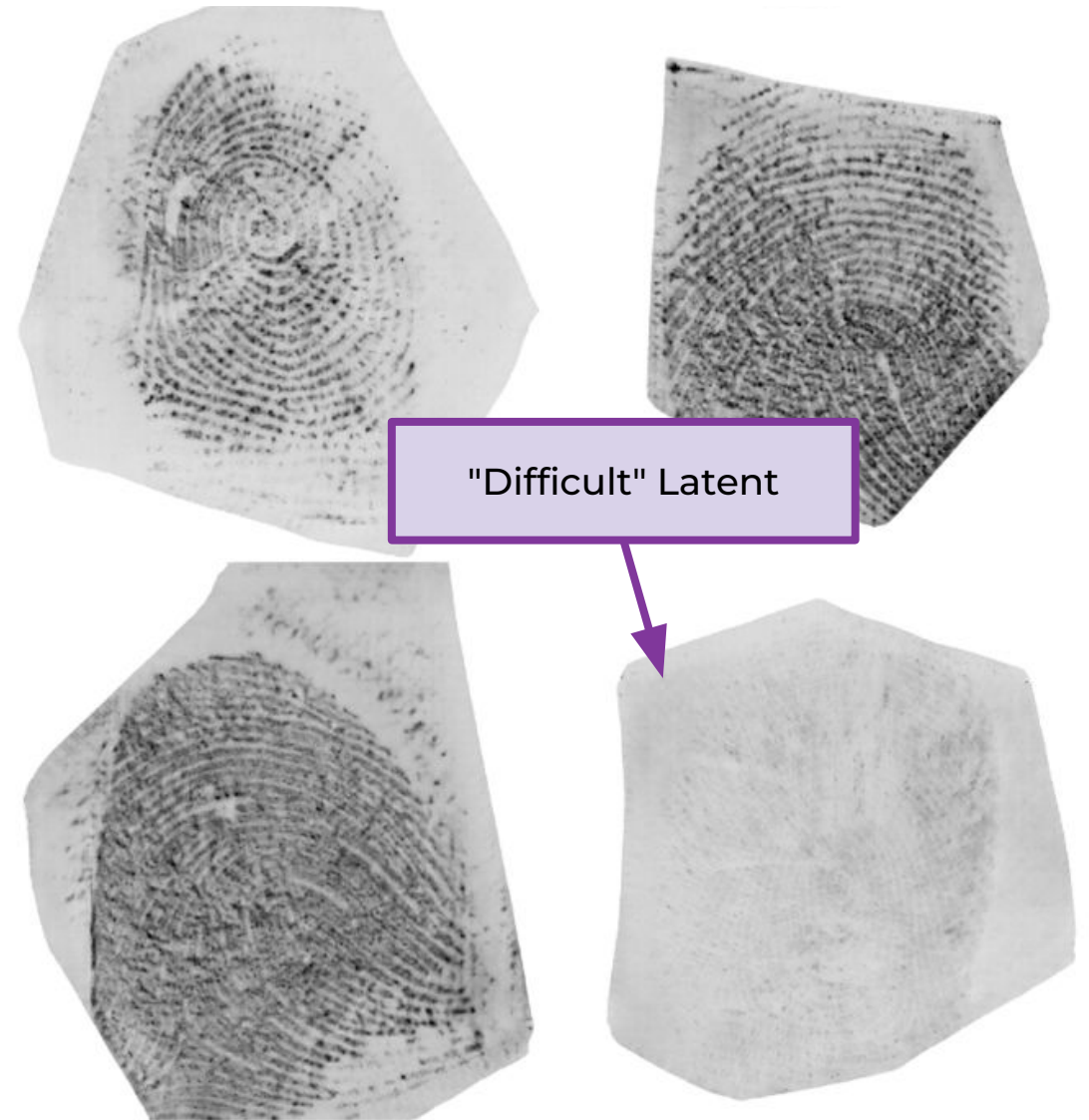
- innovatrics-0005
- Not great, not terrible results



Nutrition Report, Detection Error Tradeoff (DET)



Nutrition Report, Cumulative Match Characteristic (CMC)



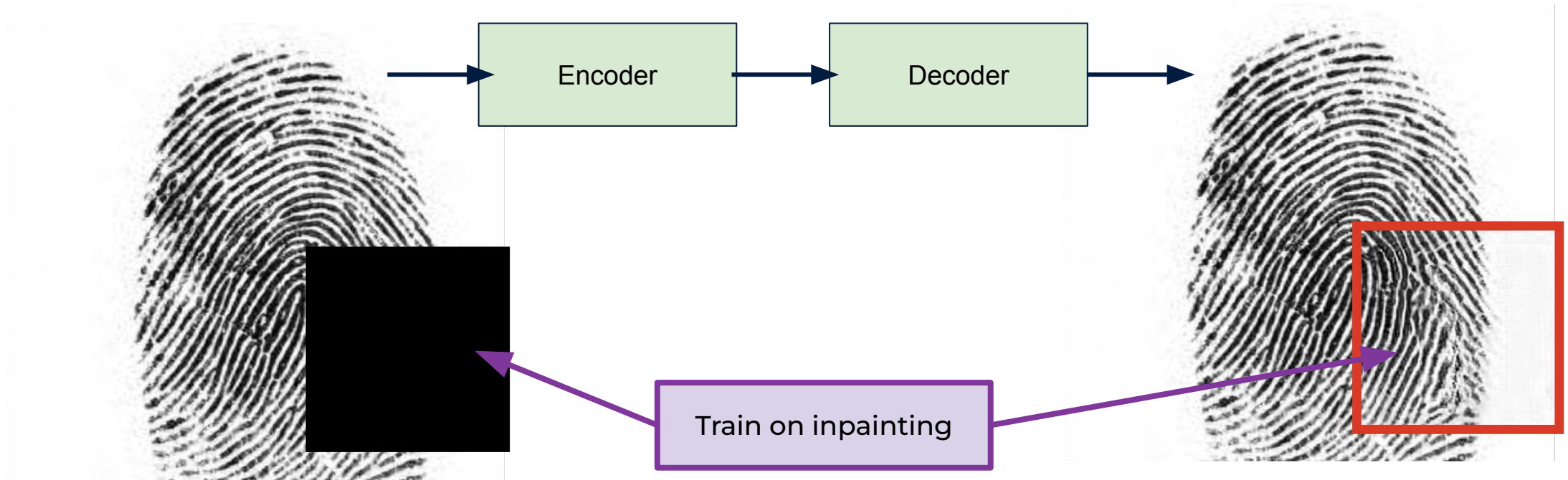
Submission 2



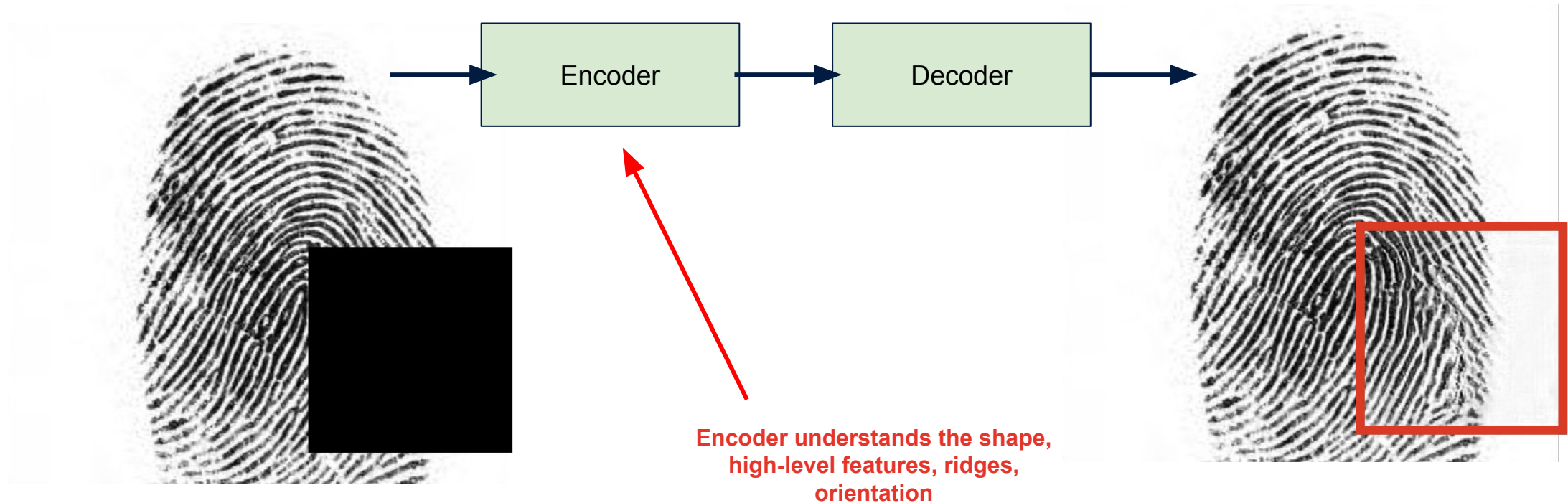
Our Submission 2

- Preserve discriminative information
- Be creative where little information is stored
- Avoid hard-coded clipping mask

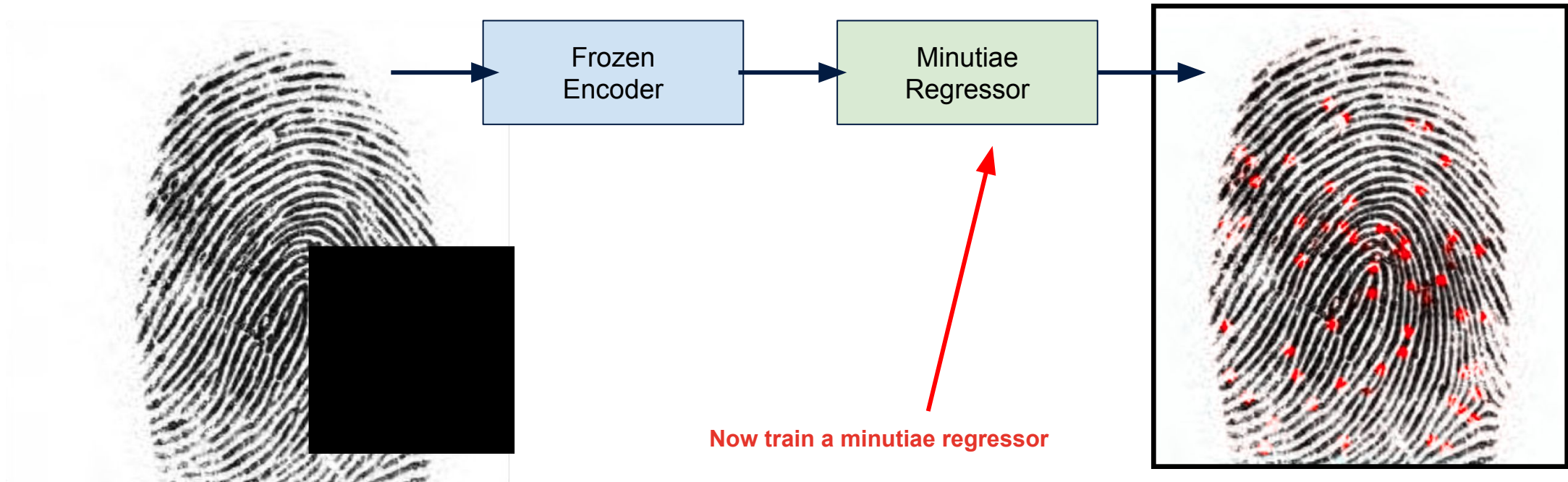
Our Submission 2

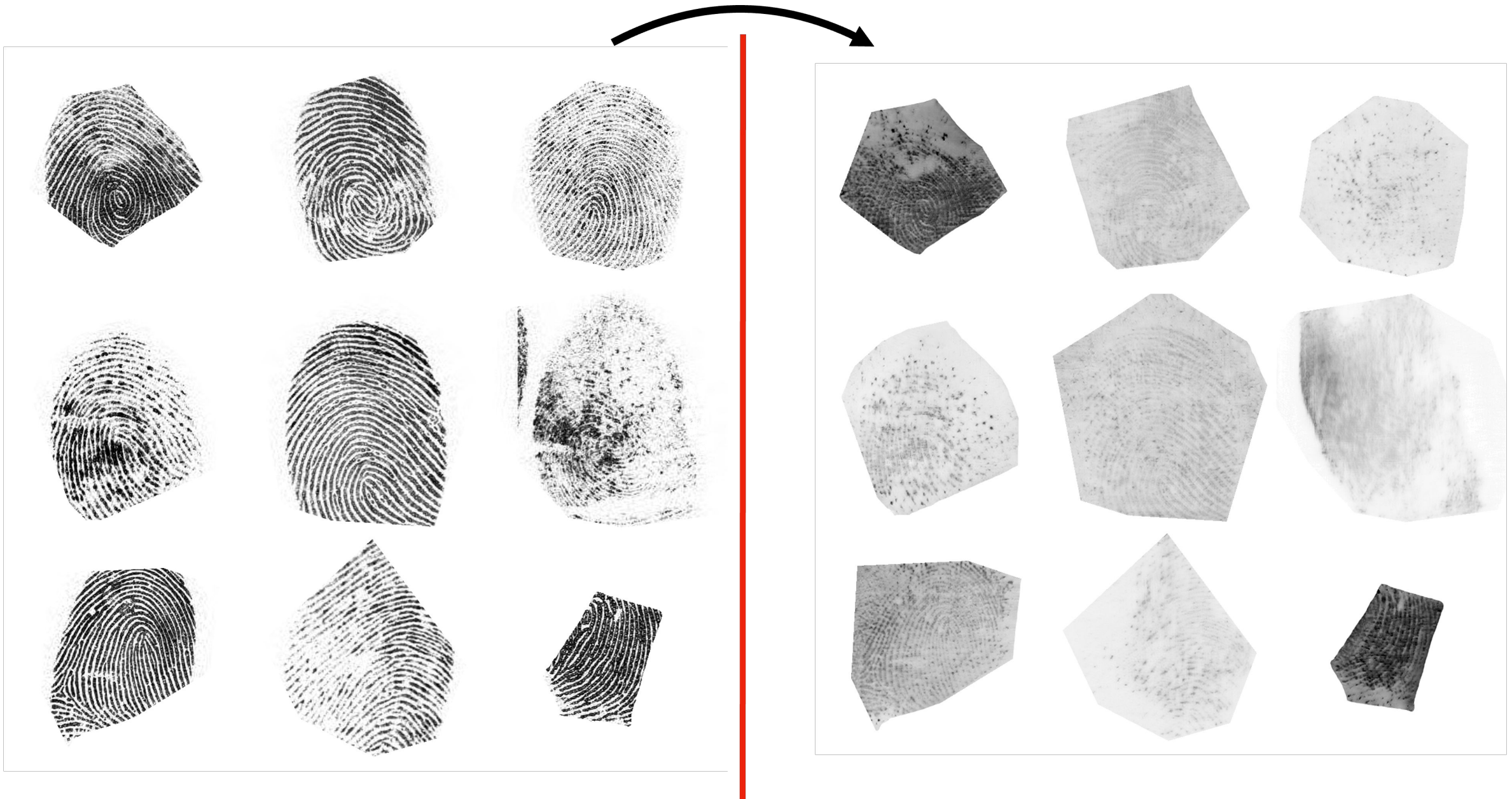


Our Submission 2



Our Submission 2

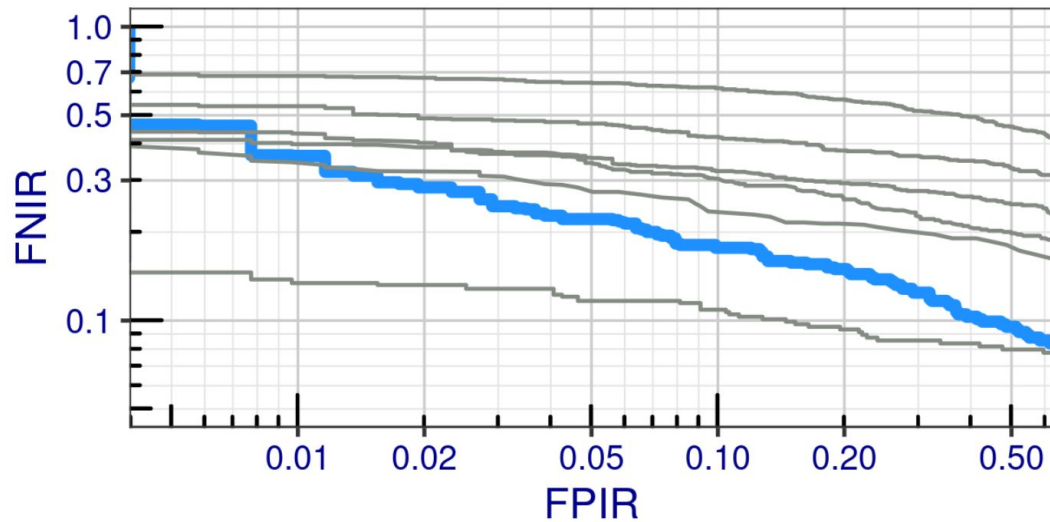




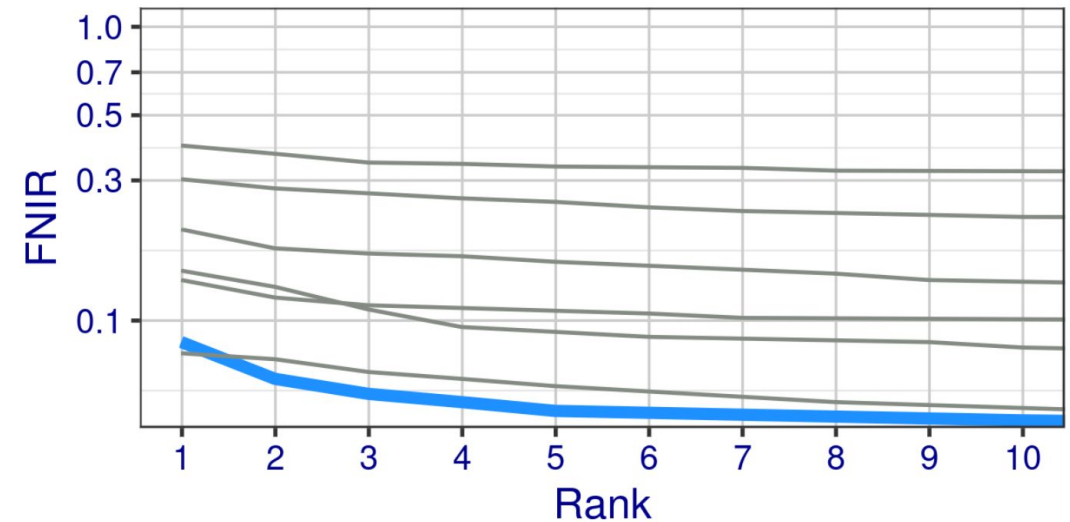


Our Submission 2

- innovatrics-0009
- Kind-of promising

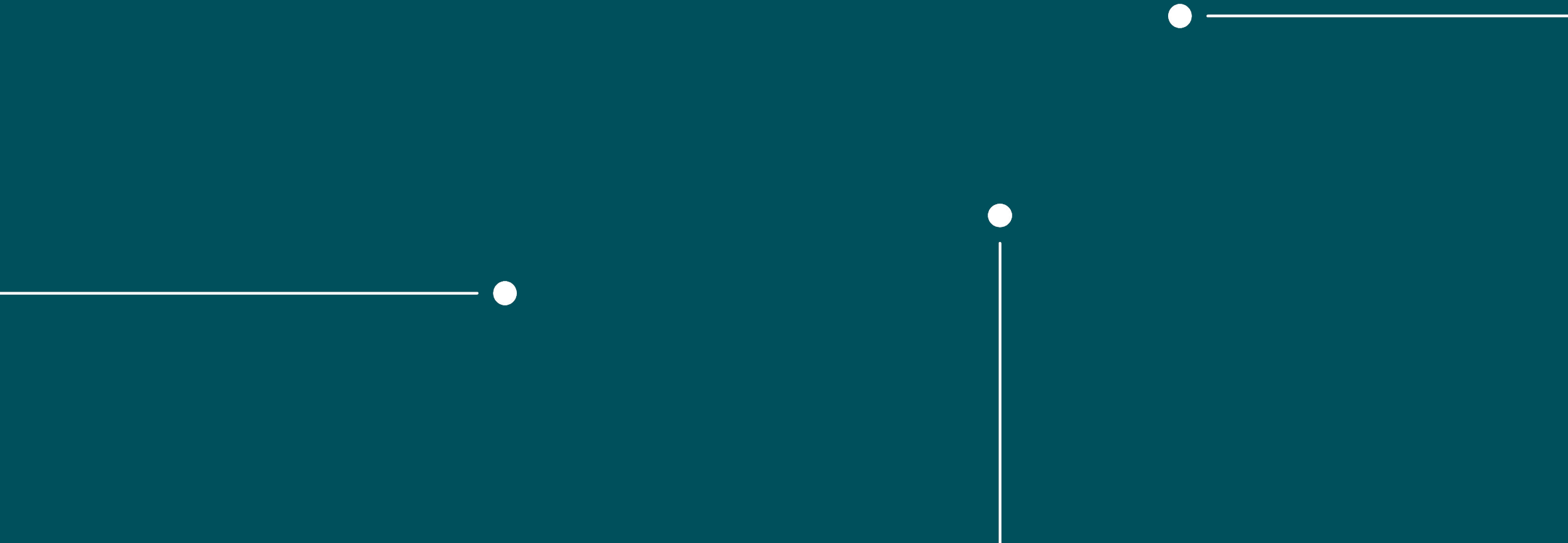


Nutrition Report, Detection Error Tradeoff (DET)



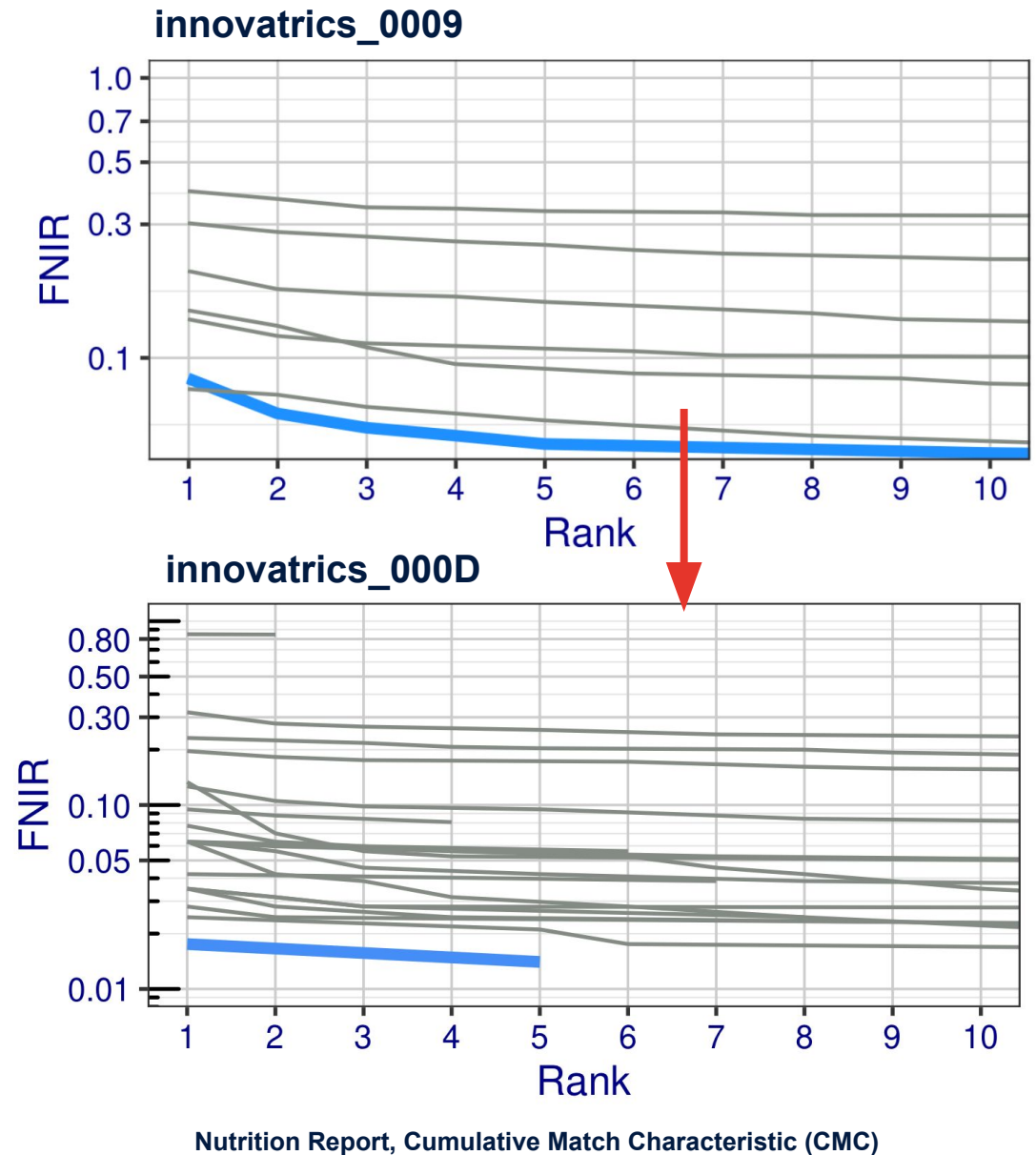
Nutrition Report, Cumulative Match Characteristic (CMC)

Current State



Current State

- Improved techniques of data synthesis
- **Still trained on purely synthetic data**



Thank you

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